

Mario Hernandez

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EDUCATION

California State University, San Marcos

Expected Graduation: May 2026

Bachelor of Science in Electrical Engineering

SKILLS

Hardware/PCB: Schematic capture, PCB layout, power design, mixed-signal systems

Embedded Systems: STM32, GPIO, ADC, PWM, SPI/I²C, SWD debugging

RF/Wireless: BLE, impedance matching, antenna basics, S-parameters

Programming & Tools: C/C++, MATLAB, Cadence Virtuoso, LTspice, HFSS

Debugging & Test: Oscilloscope, multimeter, power supply, VNA, spectrum analyzer, root cause analysis

PROJECTS

Hardware Engineering Lead – *Intra-Oral BLE Pressure Sensing Hardware Platform* Jan. 2025- Nov. 2025

- Designed a wearable Bluetooth-based pressure sensing system to measure tongue force distribution for assistive communication and medical applications
- Designed and implemented a 2-layer flexible PCB integrating 7 FSR sensors, 5V→3.3V regulation, and FPC-based programming for a wearable sensing system
- Developed analog front-end using voltage dividers to interface FSR outputs with the nRF52840 ADC for real-time data acquisition
- Diagnosed unstable sensor readings caused by high-impedance noise ($>1\text{ M}\Omega$) using oscilloscope analysis; implemented RC filtering and optimized resistor values ($1\text{ M}\Omega \rightarrow 500\text{ k}\Omega$) to improve signal stability
- Identified voltage drop across FPC power delivery; redesigned architecture to transmit 5V and regulate locally, restoring reliable operation
- Performed power rail verification, current monitoring, and signal integrity testing during hardware bring-up
- Designed STM32-based programming module for SWD programming and debugging of the nRF52840

Lead Analog Hardware Design Engineer – *Frequency-Compensated CMOS Op-Amp*

Nov. 2025

- Designed a two-stage CMOS operational amplifier with Miller compensation for stable high-gain operation in the MHz bandwidth range
- Achieved $>60\text{ dB}$ DC gain and $\sim 74^\circ$ phase margin through transistor sizing optimization and compensation capacitance sweep ($0.1\text{--}2\text{ pF}$)
- Optimized dominant pole placement and unity-gain bandwidth to ensure closed-loop stability
- Performed full circuit verification including DC operating point analysis, 4-corner PVT validation, and Monte Carlo mismatch simulation
- Implemented custom layout with matched differential pair symmetry, achieving DRC-clean and LVS-verified implementation

Embedded Systems Engineer – *PIR Motion Sensor Alarm System*

Mar. 2024

- Integrated a PIR motion sensor using interrupt-driven GPIO inputs on an STM32 Nucleo-64 board for real-time motion detection
- Developed PWM-based audio control to drive an 8-ohm speaker and generate alternating alarm tones
- Implemented LED indicators and pushbutton interrupt handling for system activation and deactivation
- Performed staged hardware bring-up and signal validation using an oscilloscope and multimeter to verify GPIO timing and PWM waveform integrity

CERTIFICATIONS

5G Introductory Level Certificate (Qualcomm)

Oct. 2024

Radio Frequency Fundamentals Workshop

Jan. 2026

PCB Basic Design Course Certificate (Altium Education)

Mar. 2026